Amey Mishra

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Skills

- Team leadership
- Project management
- Community outreach
- Robotics programming
- Engineering design
- Strategic planning

Education And Training

Expected in 05/2027
High School Diploma:
Harmony School Of Advancement
Houston

- 9th, 10th 11th grade - Honor Roll
- FIRST Tech Challenge Deans List Semifinalist
- Human Geography, CSP, CSA, World History, Statistics, Biology, Seminar, Language, Networking, Calculus BC AP Courses
- 4.75 GPA
- Member of National Honor Society, 2025, 2026
- Student Council President 9th and 10th grade
- President of 3 Clubs, Captain of 3 Teams
- Research Experience with Space Debris and their impact, Published in the American Journal of Student Research

Certifications

- Autodesk Fusion 360 Certification
- OpenEDG Python Certification
- Microsoft ITS Python Certification
- Adobe Photoshop Certification
- Adobe Illustrator Certification
- W3 schools JavaScript Certification
- FAA Certified Recreational Drone Pilot

Languages

English:
Native/ Bilingual
Hindi:

Summary

Dedicated and motivated high school student with a passion for leadership, robotics, and scientific exploration. Seeking opportunities to contribute my skills and enthusiasm to a dynamic team or project. Aspiring to pursue Aerospace and Aeronautical Engineering and wish to graduate with a Masters Degree from MIT.

Experience

Cosmobots Robotics - Founder And CEO
Houston, TX
06/2024 - Current

- Directed community outreach programs to increase STEM opportunities in underrepresented areas.
- Coordinated food, toy, and clothing drives to support local families.
- Managed two robotics teams along with various STEM competition teams.
- Secured over 20000 in funding for diverse teams within organization.
- Accumulated over 1,000 total volunteer hours, with 350 hours personally contributed.
  Dedicated 350 personal hours to various community service initiatives.

FTC 25679 The Cosmobots - Founder And Team Captain
Houston, TX
06/2023 - Current

- Led FIRST Tech Challenge 8790 as captain, achieving league finalist and regional runner-up status.
- League Connect Award Winner 3x
- Regionals Qualifier 3x
- Founded and captained FTC 25679, serving as mechanical lead since 2024.
- Functioned as mechanical lead with extensive software experience.
- Utilized effective design strategies for effective actuator use, worked within specific constraints to design a robot that could effectively solve the challenge, and created complex hardware mechanisms to solve challenges
- Used adaptive inverse kinematics, custom path planner algorithms, experimented with Square root PID, and created a custom library to implment mechanisms in robot code.
- Ranked in top 0.5% of over 8000 global teams; secured first place in Houston event, won league championship and made regional playoffs; participated in Michiana International Premier Playoffs.
- Achieved finalist position at Carolinas International Premier event.

Harmony Advancement Drone Club - Captain And President
Houston, TX
08/2023 - Current

- Led RADC Drone Competition Team 60398A in eighth grade and Team 96610A in ninth through eleventh grades.
- Managed school club with two teams, securing parts and completing competition registrations.
- Winning Alliance at Regionals, with 3 All around Champion awards, qualifying to National Championship 3 times
- Won Flight Operations Award at National Championship.
- Gained Experience with obstacle course navigation using drones, built effective presentation skills, and used python libraries and sensor feedback loops to create adaptive autonomous programs

Full Professional

Spanish:

Limited

Accomplishments

- NASA Global Space Apps Challenge Global Nominee
- NASA Space Center U program Gene Kranz Scholarship Recipient and Thermodynamics event winner, Cryogenics event runner-up
- FIRST Carolinas Premier Event Finalist Alliance and Transformation Triumph Award
- FIRST Michiana Premier Event 4th Alliance
- FIRST Houston Johnson League Championship Winning Alliance
- FIRST Houston Regional Championship Connect Award Runner-Up
- RADC Drone Houston Regional Championship 3x All Around Champion/Excellence Award
- RADC Drone Houston Regional Championship Winning Alliance
- RADC South Central National Championship Flight Operations Award
- Ecybermission Texas State Second Place
- International Math Challenge Gold and Silver Medalist
- Science Olympiad Regionals 6x Gold 1x Silver 4x Bronze
- Science Olympiad States 1x Gold 2x Bronze
- Harmony Advancement Rubiks Cube Competition 2x finalist, 2x winner
- Seaperch ROV Challenge Regional Runner-Up
- AP Scholar with Distinction Award

Volunteer Experience

- 3 years Volunteering for Harmony School of Excellence FIRST Robotics, Science Olympiad, and Drone Teams
- STEM Education in underrepresented areas through Cosmobots Robotics
- Harmony Camp Invention Volunteer (3 years), 1 week summer program for Elementary and Middle School Students

Harmony Science Olympiad Team - President
Houston, Texas
08/2024 - Current

- President of Science Olympiad Team for 2 years (Grades 9 & 10).
- 1st Place State Win: Storm the Castle (8th Grade).
- 7x Regional 1st Place Finishes in building and engineering events:
  Tower (2x) (9th & 10th Grade)
  Air Trajectory (2x) (9th & 10th Grade)
  Flight (9th Grade)
  Helicopter (10th Grade)
  Electric Vehicle (10th Grade)

Projects

AI Humanoid Robot Research and Development

- This project involved designing and programming a humanoid robot that blends physical motion with intelligent behavior. I handled the full stack: using advanced Arduino for the low-level motor control to create realistic movement, integrating OpenCV to give the robot real-time visual perception, and leveraging the OpenAI API to power complex conversational abilities and dynamic, non-scripted task planning.

ORCA - Orbital Recycling and Construction Array

- Project for the 2025 Nasa Global Space Apps Challenge. Worked on a custom neural network to analyze debris data and utilize orbital mechanics to plan paths to capture and recycle debris. Used high level CAD software to design and simulate mechanisms, and designed a full business plan for real world implementation

EcoScape Sustainable Architecture

- Researched, designed, and planned a sustainable city based on current technologies. Utilized environmental data, analyzed current ecological issues, and made configurations for a variety of climates. Texas State Finalist in the 2024 Ecybermission

Radio-Controlled 3D Printed Airplane for aerodynamics testing and data collection

- CAD designed and 3D printed a small aircraft style RC airplane in order to gain basic flight experience, gather aerodynamic data, and test implementation of flaps, spoilers, and other control surfaces

3D printed drone for location mapping with waypoint navigation

- CAD designed and 3D printed a UAV to navigate waypoints along set coordinates. Implemented a high resolution camera with a video downlink in order to ensure safety and continuous data collection

Using AI and neural networks to search through NASA exoplanet data in order to analyze celestial bodies

- Used NASA public datasets and trained neural networks to identify possible candidates for Exoplanets
- Improved accuracy and consistency of a custom CNN (Convolutional Neural Network)

Activities And Honors

- Participation in VEX Robotics Competition (Captain)
- NASA Global Conrad Challenge Participant (Awaiting Results)
- Samsung Solve for Tomorrow Participant (Awaiting Results)
- Participation in SkillsUSA Engineering Competition
- Participation in SeaPerch Underwater Robotics Competition (Captain/Club President)
- Participation in official WCA Rubiks Cube Competition
- Participation in Cyberpatriot Programming Competition

References

References available upon request.